

Fingerprint Liveness Detection (Presentation Attack Detection)

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Abstract

This project implements a Fingerprint Presentation Attack Detection (PAD) system using MobileNetV3-Small with transfer learning. The model classifies fingerprint images as LIVE or SPOOF using the SOCOFing dataset and synthetic spoof samples.

Introduction

The objective is to detect presentation attacks before fingerprint authentication. A lightweight transfer learning approach was implemented using PyTorch in Google Colab.

Dataset

SOCOFing dataset was used.

LIVE Images: 60

SPOOF Images: 60

Split:

- Train: 42 + 42
- Validation: 9 + 9
- Test: 9 + 9

Methodology

- Resize to 224×224
- Data augmentation for training
- ImageNet normalization
- MobileNetV3-Small transfer learning
- Adam optimizer
- CrossEntropyLoss
- 20 epochs

Results

Best Validation Accuracy: 100%
Validation Samples: 18
Equal Error Rate (EER): 0.0
Overall Test Accuracy: 16/18 (88.89%)

Evaluation Plots

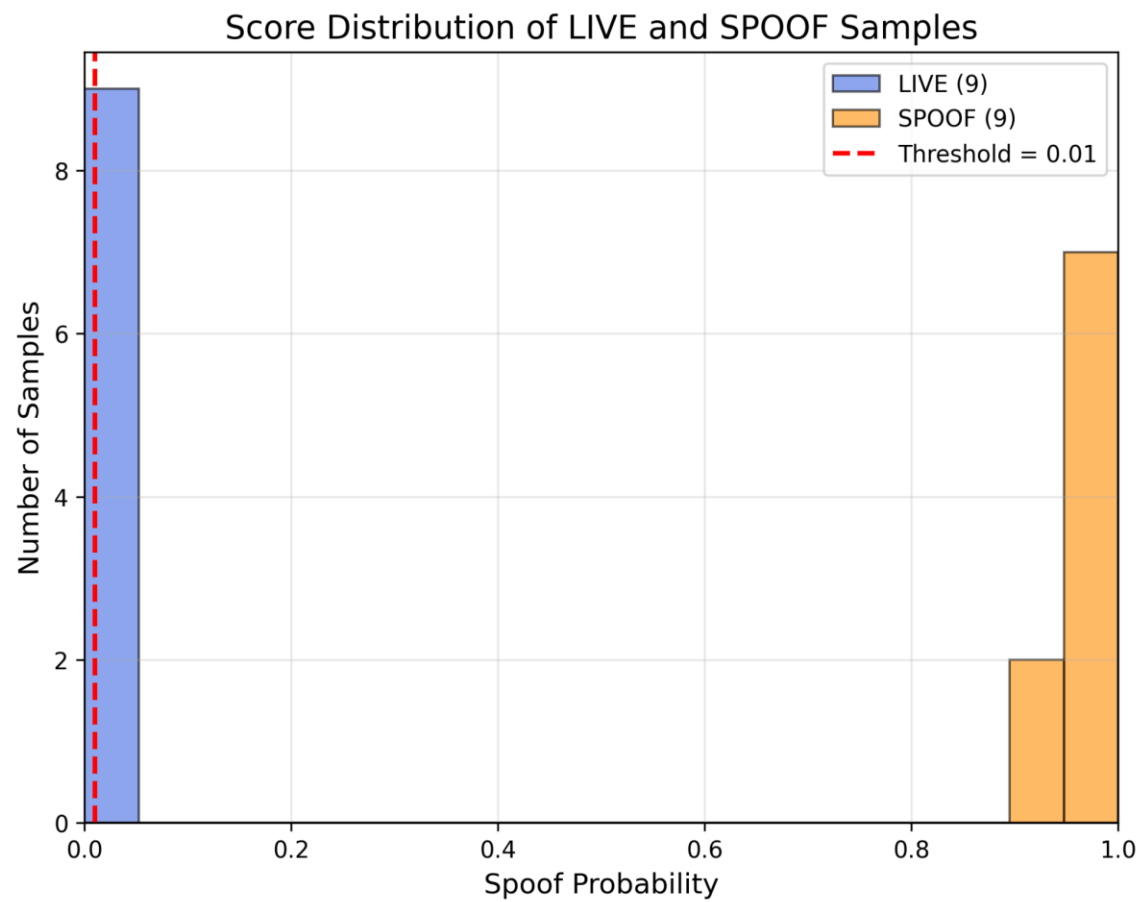


Figure 1. Score Distribution of LIVE and SPOOF Samples

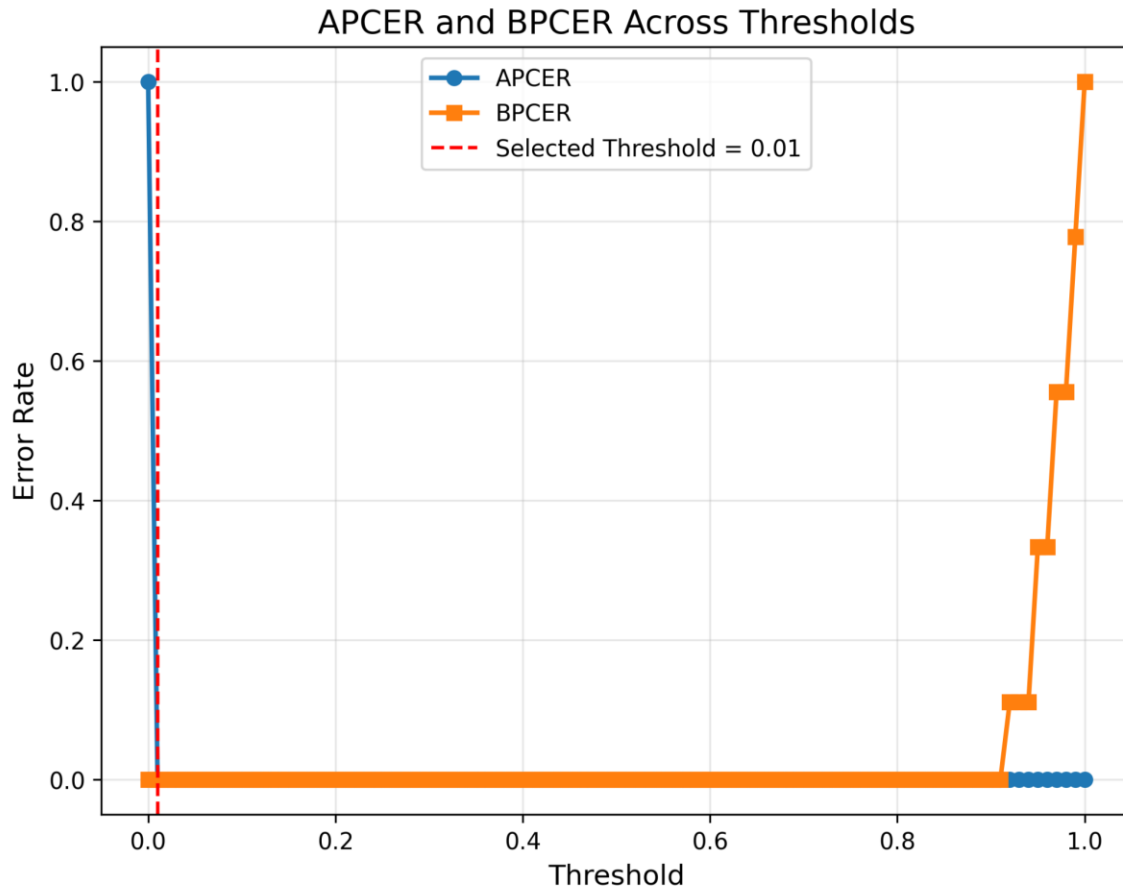


Figure 2. APCER and BPCER Across Thresholds

Assignment Discussion

Q1. Dataset: SOCOFing with 60 LIVE and 60 SPOOF images.

Q2. Backbone: MobileNetV3-Small was selected because it is lightweight, efficient and suitable for deployment while maintaining good accuracy.

Q3. Threshold: Validation selected an operating threshold close to 0.01. Due to only 18 validation samples, BPCER could not be fixed exactly at 3%; the nearest operating point was used.

Q4. EER: The validation Equal Error Rate (EER) was 0.0.

Q5. Hardest attacks: High-quality silicone or 3D mould attacks are generally the hardest because they closely resemble genuine fingerprints.

Q6. Why fix BPCER? It limits false rejection of genuine users while measuring security

through APCER.

Q7. Future work: Use larger datasets with real spoof fingerprints, improve threshold calibration, evaluate on benchmark PAD datasets and optimize for mobile deployment.

Limitations

This work used a relatively small dataset with synthetic spoof images. Larger datasets containing real presentation attacks would improve robustness and generalization.

Conclusion

The proposed MobileNetV3-Small based PAD system successfully distinguishes LIVE and SPOOF fingerprint images using transfer learning and demonstrates promising performance for lightweight biometric security applications.